

Adding the new SI prefixes

Document #: P2734R0
Date: November 30, 2022
Project: Programming Language C++
Library Group
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Abstract

We propose to add the missing SI prefixes **quecto** (10^{-30}), **ronto** (10^{-27}), as well as **ronna** (10^{27}) and **quetta** (10^{30}) to the `<ratio>` header.

1 Motivation and Scope

The General Conference on Weights and Measures (CGPM), at its 27th meeting in November 2022, decided [[CGPM 2022 Resolution 3](#)]

...to add to the list of SI prefixes to be used for multiples and submultiples of units the following prefixes:

Multiplying factor	Name	Symbol
10^{27}	ronna	R
10^{-27}	ronto	r
10^{30}	quetta	Q
10^{-30}	quecto	q

This decision directly affects [[ratio.syn](#)] and [[ratio.si](#)], which contain **ratio** typedefs for *each* SI prefix. If the list of SI prefixes grows (it last did in 1991), the corresponding list of **ratio** typedefs needs to follow suit.

2 Implementability & Impact on the Standard

The multiplying factors denominated by the new SI prefixes cannot be represented as a **ratio** when `intmax_t` is 64-bit. It is a property they share with the existing prefixes **yocto**, **zepto**, **zetta** and **yotta**, which are therefore optional in the current IS.

Platforms that can represent 10^{24} (**yotta**) in `intmax_t` can also likely represent 10^{30} (**quetta**) in it (if `intmax_t` is 128-bit ($10^{30} < 2^{100}$)). Even if `intmax_t` is an 80- or 96-bit entity, the problem, from a wording point of view, remains the same: Non-representable typedefs are not required to be provided.

Such proposals, however, would be much more demanding on committee time and require LEWG involvement, so this proposal steers clear of such desires and stays within the existing wording for **yotta** etc to deliver the missing SI prefixes with as little effort as possible.

The following is relative to [N4917]:

- ```
#define __cpp_lib_ratio YYYYMMML // also in <ratio>
```

- ```
// [ratio.si], convenience SI typedefs
+ using quecto = ratio<1, 1'000'000'000'000'000'000'000'000'000>; // see below
+ using ronto  = ratio<1,      1'000'000'000'000'000'000'000'000'000>; // see below
using yocto    = ratio<1,          1'000'000'000'000'000'000'000'000'000>; // see below
using zepto    = ratio<1,            1'000'000'000'000'000'000'000'000'000>; // see below
[...]
using zetta    = ratio<              1'000'000'000'000'000'000'000'000, 1>; // see below
using yotta    = ratio<              1'000'000'000'000'000'000'000'000, 1>; // see below
+ using ronna   = ratio<        1'000'000'000'000'000'000'000'000'000, 1>; // see below
+ using quetta  = ratio<1'000'000'000'000'000'000'000'000'000'000, 1>; // see below
```

- Change **[ratio.si]** as indicated:

4 References

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- ¹At least, this author is not aware of any such proposals at the time of writing